

AI for Financial Practitioners

Lecture 1 · Part II

From Intelligence to a Governed Operating System

From impressive answers to dependable institutional work

The demonstration is not the service

SCALE CHANGES THE PROBLEM

One impressive answer

- Human remains close to the work
- Ambiguity is resolved informally
- Errors are noticed before action
- Context is reconstructed on demand



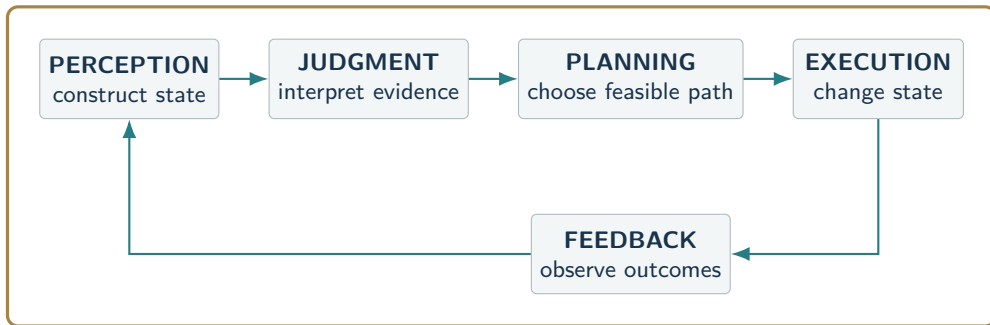
Continuous institutional service

- Information arrives asynchronously
- State and evidence become stale
- Failures interact across components
- Outputs may trigger consequences

The governed intelligence cycle

AN OPERATING GRAMMAR FOR INSTITUTIONAL AI

GOVERNANCE SURROUNDS EVERY TRANSITION



Perception: construct the right state before reasoning

CONTEXT IS A CONTROL, NOT DECORATION

Identity · issuer, security, counterparty

Time · event, effective, ingestion, current

Definition · units, accounting, legal terms

Authority · approved source and version

Completeness · known, missing, contradictory

Dangerous institutional illusion

A system can produce a visually complete answer while its underlying state is incomplete.

- blank → zero
- stale value → “current”
- missing fact → plausible prose

Judgment and planning: intelligence must remain heterogeneous

INTERPRET FIRST; THEN CHOOSE A FEASIBLE ACTION

JUDGMENT

Retrieved facts provide evidence.

Deterministic calculations provide exact quantities.

Predictive models estimate uncertain outcomes.

Generative reasoning integrates context and competing explanations.

PLANNING

Objective — what are we trying to achieve?

Constraints — mandate, limits, liquidity, law.

Alternatives — no action, analyze, escalate, transact.

Dependencies — what must be revalidated before action?

Execution: consequences change the standard

CAPABILITY DOES NOT CONFER AUTHORITY

Execution controls

- Fresh authorization at the action boundary
- Least-privilege tool permissions
- Confirmation and unique action identifiers
- Idempotence against duplicate actions
- Reconciliation of intended vs. actual state

Delegation should tighten as reversibility falls



Stronger evidence, narrower limits and more explicit confirmation should accompany less reversible actions.

Feedback is not automatically learning

OUTCOMES REQUIRE INTERPRETATION BEFORE ADAPTATION



A correction may indicate

- factual error
- changed policy
- reviewer preference
- model/retrieval failure

An outcome may reflect

- decision quality
- luck / market movement
- execution quality
- changed external conditions

Governance is embedded in the operating cycle

CONTROL THE TRANSITIONS, NOT ONLY THE MODEL

Transition		Executive question	Control examples
Evidence State	→	What may the system treat as true now?	lineage, freshness, authorization, abstention
State → Judg- ment		What supports the interpretation?	citations, calculations, counterevidence, uncertainty
Judgment → Plan		Is the proposed path feasible?	mandate, limits, optimization, escalation
Plan → Action		Who may cause the consequence?	permissions, approval, confirmation, reversibility
Outcome Change	→	What are we allowed to learn?	incident review, validation, change authority

Operating artifact: the AI service charter

MAKE THE SYSTEM GOVERNABLE IN ONE PAGE

1 · PURPOSE & STATE

responsibility · user ·
authoritative inputs · update
expectations · abstention 2 ·

AUTHORITY

tools · data classes · actions ·
limits · confirmations ·
escalation owners

3 · ASSURANCE

acceptance criteria · baseline ·
evidence · numerical correctness
· reviewer burden 4 ·

MONITORING

health · quality · control ·
economics · thresholds ·
graceful degradation

5 · INCIDENTS

stop · isolate · reconstruct ·
communicate · restore 6 ·

CHANGE

ownership of models · prompts
· retrieval · code · policy ·
permissions

From AI capability to institutional intelligence

THE EXECUTIVE OPERATING MODEL

PERCEIVE · construct a valid, time-aware state

JUDGE · separate evidence, calculation and interpretation

PLAN · optimize under objectives and constraints

EXECUTE · act only within explicit authority

FEEDBACK · measure outcomes without automatic learning

GOVERN · monitor, intervene, degrade and recover

The unit of analysis changes

Not the model.

Not the prompt.

Not even the agent.

The institution.

Its intelligence emerges from coordinated **data, models, tools, policies and people.**